

## 2.1 Percentage increase

The price of a designer T-shirt was \$85.

The shop increases the price by 20%.

To calculate the new price, we increase \$85 by 20%.

When a quantity is increased by a given percentage, it is made larger. The percentage amount is **added** to the original quantity.

**EXAMPLE**

**Increase \$85 by 20%.**

**Method 1:**

$$\begin{aligned} 20\% \text{ of } \$85 &= \frac{20}{100} \times 85 \\ &= \$17 \end{aligned}$$

$$\begin{aligned} \text{The increased amount is } & \$85 + \$17 \\ &= \$102 \end{aligned}$$

**Method 2:**

$$\begin{aligned} \text{The new amount is } & 100\% + 20\% \\ &= 120\% \text{ of the original} \\ 120\% \text{ of } \$85 &= \frac{120}{100} \times 85 \\ &= \$102 \end{aligned}$$

**1** Increase each amount by 20%.

**a** \$60

**b** \$150

**c** \$230

**d** 750 mL

**e** 80 kg

**f** 4.8 metres

**2** Complete these percentage increase calculations.

**a** Increase \$50 by 10%.

**b** Increase \$200 by 12%.

## 2.1 Percentage increase

CONTINUED

**c** Increase \$42 by 30%.

**d** Increase \$168 by 25%.

**e** Increase \$690 by 15%.

**f** Increase \$28.50 by 10%.

**g** Increase 80 km by 40%.

**h** Increase 60 tonnes by 5%.

**i** Increase 3.2 kg by 15%.

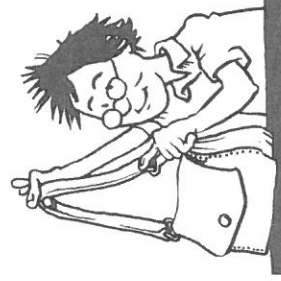
**j** Increase \$120 by 12.5%.

**Answer the following questions in your workbook.**

**3** A pair of sports shoes is priced at \$130.  
What is the new price if the shop increases all prices by 15%?

**4** A satchel cost \$72 to make.  
Calculate the selling price if the cost is marked up by 40%.

**5** Jamal's current weekly pay is \$740.  
His pay is increased by 3%.  
What is his new weekly pay?



## 2.2 Percentage decrease

A skateboard is priced at \$230.

During a sale, the price is reduced by 10%.

To calculate the sale price, we **decrease \$230 by 10%**.

When a quantity is decreased by a given percentage, it is made smaller. The percentage amount is **subtracted** from the original quantity.

**EXAMPLE**

Decrease \$230 by 10%.

**Method 1:**

$$10\% \text{ of } \$230 = \frac{10}{100} \times 230 \\ = \$23$$

$$\text{The decreased amount is } \$230 - \$23 \\ = \$207$$

**Method 2:**

$$\begin{aligned} \text{The new amount is } 100\% - 10\% \\ = 90\% \text{ of the original} \\ 90\% \text{ of } \$230 &= \frac{90}{100} \times 230 \\ &= \$207 \end{aligned}$$



1 Decrease each amount by 10%.

a \$80

b \$130

c \$65

d 200 km

e 450 litres

f 9.8 kg

2 Complete these percentage decrease calculations.

a Decrease \$200 by 12%.

b Decrease \$120 by 15%.

## 2.2 Percentage decrease

CONTINUED

c Decrease \$340 by 25%.

d Decrease \$1 570 by 40%.

e Decrease \$645 by 20%.

f Decrease \$18 by 5%.

g Decrease 120 metres by 45%.

h Decrease 160 kg by 20%.

i Decrease 70 litres by 15%.

j Decrease 200 km by 12.5%.

Answer the following questions in your workbook.

3 The population of a town was 18 400. It has decreased by 8%. What is the current population of this town?

4 A white-water rafting weekend is priced at \$1 850. During a sale, the price is reduced by 30%. What is the sale price?

5 Katie's weekly wage is \$835. This is decreased by 25% when income tax is deducted. How much remains?

